

AI Homework 2 Report - Image-to-Video Semantic Retrieval via Object Detection

I imported a pre-trained YOLO model and fine-tuned it on the provided training dataset. I chose YOLO because I've worked with this model before, and it's a generally reliable computer vision model.

My method for extracting frames from the video involved sampling a frame every three seconds. I came up with this interval by watching and observing roughly how long the director holds a shot for, and realized it's generally at least 3 seconds. This also prevents the timestamps from being overly fragmented, as every time period will have at least a 3 second space.

My main data structure for the semantic search and frame parsing was a Pandas DataFrame. The columns 0-22 were each an exterior car part, while the 23rd column was a list of every filename from the extracted video frames. This DataFrame was populated with a 1 under each element appearing in that frame, and a 0 otherwise. The first 10 entries are displayed below.

back_bumper	back_door	back_glass	back_left_door	back_left_light	back_light	back_right_door	back_right_light	front_bumper	front_door	front_glass	front_left_door	front_left_light	front_light	front_right_door	front_right_light	hood	left_mirror	object	right_mirror	tailgate	trunk	wheel	path
0	0	0	0	0	0	0	0	0	0	1	0	0	0	1	0	0	0	0	0	0	0	1	frame_0001.jpg
0	0	0	0	0	0	0	0	1	0	1	0	0	0	0	0	1	1	0	1	0	0	1	frame_0002.jpg
1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	1	frame_0003.jpg
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	frame_0004.jpg
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	frame_0005.jpg
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	frame_0006.jpg
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	frame_0007.jpg
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	frame_0008.jpg
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	frame_0009.jpg
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	frame_0010.jpg

I parsed through the frame column of this DataFrame and ran the predictor model on it. Extracting the information from boxes, I was able to populate this DataFrame, as well as the output DataFrame that I eventually convert to a parquet file.

My semantic search is set up to where you can parse any model and any input image. First the model makes detections on the input image. For each element detected, it iterates through the DataFrame I set up and looks for matches. It does this per element, and drops any repeated elements detected in the image. Originally, I failed to drop repeats, and it led to overlaps and redundancies in the output table. The index of the row corresponds to the frame. I use the index and multiply by 3 to extract the number of seconds in the video.

AI Homework 2 Report - Image-to-Video Semantic Retrieval via Object Detection

I set a flag named 'started' that is False. When there is a match in the DataFrame, the start_time is recorded and the flag is set to True. Once the loop reaches a row where the element is no longer being detected, the end_time is recorded and the flag is set back to False. There is an output table that will then record the element, start_time and end_time. The time is recorded using the built in python time library, where the seconds are converted to a string that is the time stamp. The total number of elements is determined by the number of rows.

	class_label	start_time	end_time
0	wheel	00:00	00:09
1	wheel	02:21	02:24
2	wheel	07:45	07:48
3	wheel	18:39	18:42
4	wheel	18:57	19:00
..	...	...	...
104	front_right_door	26:57	27:00
105	front_right_door	39:03	39:06
106	front_right_door	39:39	39:45
107	front_right_door	41:36	41:42
108	front_right_door	45:42	45:45

```
[109 rows x 3 columns]
Number of supporting detections: 109
```

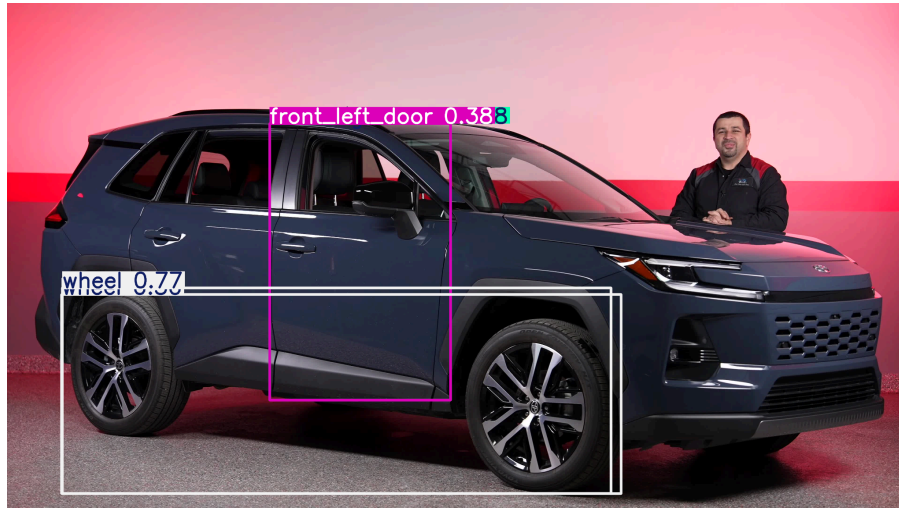
My output DataFrame held the following columns: "VID", "Frame Index", "Class Label", "Bounding Box", "Confidence Score".

Each row represents a frame. The class labels are a list of elements detected. The confidence score corresponds to this list by index. Bounding boxes do as well. This column contains a list of lists, as the x-y coordinates are recorded in sets of 4.

When compiling the DataFrame, a mistake I made was forgetting to omit the frames that had no detections, so this was a fix I had to implement. I had to add a second index in my loop and a conditional that checks if the class label list is empty before adding it to the DataFrame. I was also unfamiliar with the parquet file format, so I had some difficulties converting my DataFrame to the output format, taking Gemini's help for this.

AI Homework 2 Report - Image-to-Video Semantic Retrieval via Object Detection

In terms of failure cases, the YOLO model is not perfect. Sometimes there are overlapping detections. For example, in the following image the front door is detected as both the front right door and front left door. There are also two wheels detected, but both detections include both wheels in the bounding boxes.



Overall, the assignment was an interesting exercise that helped teach me about algorithms that implement AI models.